

TCC70xx MCU BSP

API Specification for General Purpose Input Output

Rev. 0.10 [A]

2021-09-03

Preliminary version

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1 INTRODUCTION

This document describes the general purpose input/output device driver (hereinafter referred to as GPIO driver) of TCC70xx MCU BSP. For more detailed information on general purpose input/output in the TCC70xx MCU Sub-system, refer to Chapter 23 General Purpose I/O (GPIO) and Port Multiplexing in "*TCC70xx Full Specification*". [1]

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

GPIO	General Purpose Input Output
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3 SOURCE FILES

3.1 Location of Source Files

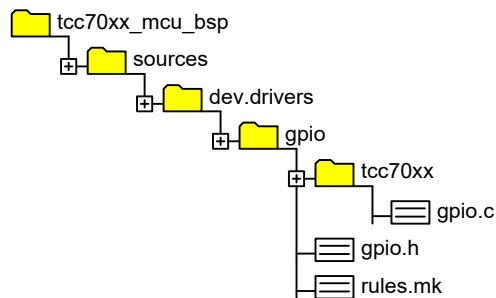


Figure 3.1 Location of Source Files

3.2 Simple Description of Source Files

- gpio.c
 - GPIO driver source
- gpio.h
 - GPIO driver API header
- rules.mk
 - Rules used for build

4 CONSTANTS

4.1 GPIO Port Configuration

Constant values for GPIO port configuration

Declaration

```
#define GPIO_INPUTBUF_SHIFT    10
#define GPIO_INPUTBUF_MASK    0x3
#define GPIO_INPUTBUF_EN      ((2 | 1) << GPIO_INPUTBUF_SHIFT)
#define GPIO_INPUTBUF_DIS     ((2 | 0) << GPIO_INPUTBUF_SHIFT)

#define GPIO_OUTPUT_SHIFT     9
#define GPIO_OUTPUT           (1 << GPIO_OUTPUT_SHIFT)
#define GPIO_INPUT            (0 << GPIO_OUTPUT_SHIFT)

#define GPIO_CD_SHIFT         6
#define GPIO_CD_MASK          0x7
#define GPIO_CD(x)            (((x) & GPIO_CD_MASK) | 0x4) << GPIO_CD_SHIFT)

#define GPIO_PULL_SHIFT       4
#define GPIO_PULL_MASK        0x3
#define GPIO_NOPULL           (0 << GPIO_PULL_SHIFT)
#define GPIO_PULLUP           (1 << GPIO_PULL_SHIFT)
#define GPIO_PULLDN           (2 << GPIO_PULL_SHIFT)

#define GPIO_FUNC_MASK        0xF
#define GPIO_FUNC(x)          ((x) & GPIO_FUNC_MASK)
```

Description

Value	Description
GPIO_INPUTBUF_SHIFT	Input buffer shift for parsing the 2nd parameter of GPIO_Config()
GPIO_INPUTBUF_MASK	Input buffer mask
GPIO_INPUTBUF_EN	Input buffer enable
GPIO_INPUTBUF_DIS	Input buffer disable
GPIO_OUTPUT_SHIFT	Output shift for parsing the 2nd parameter of GPIO_Config()
GPIO_OUTPUT	GPIO Output
GPIO_INPUT	GPIO Input
GPIO_CD_SHIFT	GPIO driver strength shift for parsing the 2nd parameter of GPIO_Config()
GPIO_CD_MASK	GPIO driver strength mask
GPIO_CD(x)	GPIO driver strength

<i>Value</i>	<i>Description</i>
<i>GPIO_PULL_SHIFT</i>	<i>GPIO pull shift for parsing the 2nd parameter of GPIO_Config()</i>
<i>GPIO_PULL_MASK</i>	<i>GPIO pull mask</i>
<i>GPIO_NOPULL</i>	<i>GPIO No pull</i>
<i>GPIO_PULLUP</i>	<i>GPIO pull up</i>
<i>GPIO_PULLDN</i>	<i>GPIO pull down</i>
<i>GPIO_FUNC_MASK</i>	<i>GPIO function mask</i>
<i>GPIO_FUNC(x)</i>	<i>GPIO function</i>

Remark

None

4.2 GPIO Port and Pin Structure

Constant values for GPIO port address

Declaration

```
#define GPIO_PORT_SHIFT    5
#define GPIO_PORT_MASK    (0x1F << GPIO_PORT_SHIFT)

#define GPIO_GPA(x)        (GPIO_PORT_A | ((x) & 0x1F))
#define GPIO_GPB(x)        (GPIO_PORT_B | ((x) & 0x1F))
#define GPIO_GPC(x)        (GPIO_PORT_C | ((x) & 0x1F))
#define GPIO_GPK(x)        (GPIO_PORT_K | ((x) & 0x1F))
```

Description

Value	Description
GPIO_PORT_SHIFT	GPIO port shift
GPIO_PORT_MASK	GPIO port mask
GPIO_GPA(x)	GPIO A port address
GPIO_GPB(x)	GPIO B port address
GPIO_GPC(x)	GPIO C port address
GPIO_GPK(x)	GPIO K port address

Remark

None

5 APIs

5.1 GPIO_Config

GPIO port configuration user interface APIs. Refer to the constants in Chapter 4.1 GPIO Port Configuration for supported configuration.

Declaration

```
SALRetCode_t GPIO_Config  
(  
    uint32 uiPort,  
    uint32 uiConfig  
);
```

Parameters

Value	Description
<i>uiPort</i>	<i>[in] GPIO port number</i>
<i>uiConfig</i>	<i>[in] GPIO port configuration</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

5.2 GPIO_Get

Gets the value of the GPIO port corresponding to the input parameter.

Declaration

```
uint8 GPIO_Get  
(  
    uint32 uiPort  
);
```

Parameters

Value	Description
<i>uiPort</i>	<i>[in] GPIO port number</i>

Return

Value	Description
<i>0</i>	<i>GPIO port value low</i>
<i>1</i>	<i>GPIO port value high</i>

Remark

None

5.3 GPIO_Set

Sets the 2nd input parameter value to the GPIO port corresponding to the 1st input parameter.

Declaration

```
SALRetCode_t GPIO_Set  
(  
    uint32 uiPort,  
    uint32 uiData  
);
```

Parameters

Value	Description
<i>uiPort</i>	<i>[in] GPIO port number</i>
<i>uiData</i>	<i>[in] GPIO port value</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

5.4 GPIO_ToNum

Returns the GPIO port pin number.

Declaration

```
sint32 GPIO_ToNum  
(  
    uint32 uiPort  
);
```

Parameters

<i>Value</i>	<i>Description</i>
<i>uiPort</i>	<i>[in]</i> GPIO port number

Return

<i>Value</i>	<i>Description</i>
<i>Greater than 0</i>	<i>Port pin number</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

5.5 GPIO_PerichSel

Nth Peripheral Channel Selection. Refer to Chapter 23.3.3 Peripheral Channel Selection in “*TCC70xx Full Specification*”. Related peripherals: UART, I2C, GPSB, I2S, PWM

Declaration

```
SALRetCode_t GPIO_PerichSel  
(  
    uint32 uiPerichSel  
    uint32 uiCh  
);
```

Parameters

Value	Description
<i>uiPerichSel</i>	<i>[in] Index of peripheral selection</i>
<i>uiCh</i>	<i>[in] Index of peripheral channel</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

5.6 GPIO_MfioCfg

Sets Multi-Function I/O (MFIO) configuration register value. Refer to Chapter 23.3.4 Multi-Function I/O in “*TCC70xx Full Specification*”.

Related peripherals: GPSB, UART, I2C

Declaration

```
sint32 GPIO_MfioCfg
(
    uint32 uiPeriSel,
    uint32 uiPeriType,
    uint32 uiChSel,
    uint32 uiChNum
);
```

Parameters

Value	Description
<i>uiPeriSel</i>	<i>[in] Index of MFIO peripheral selection</i>
<i>uiPeriType</i>	<i>[in] Type of MFIO peripheral selection</i>
<i>uiChSel</i>	<i>[in] Index of MFIO channel selection</i>
<i>uiChNum</i>	<i>[in] Value of MFIO channel</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

6 HOW TO USE GPIO DRIVER

6.1 Example: GPIO Driver APIs

```
int main()
{
    SALRetCode_t  ret = SAL_RET_FAILED;
    uint32        value;
    uint32        pin_num;

    ret = GPIO_Config(GPIO_GPA(27UL),           // target port : GPA Port 27th pin
                      ((uint32)GPIO_FUNC((uint32)1UL) | // configuration : Function 1,
                      (uint32)GPIO_OUTPUT |           // GPIO output mode
                      GPIO_PULLDN |                  // GPIO pull down
                      GPIO_CD((uint32)0x3UL));        // GPIO driver strength

    Value        = GPIO_Get(GPIO_GPK(5));           // return the value of GPIO_K port 5 pin
    ret          = GPIO_Set(GPIO_GPK(5), 1UL);      // set 1UL value on GPIO_K port 5 pin
    pin_num      = GPIO_ToNum(GPIO_GPB(3));         // return the value of GPIO_B port 3 pin number

    Return ret;
}
```

7 REFERENCES

- [1] TCC70xx Full Specification
- [2] Contact Telechips for more details: sales@telechips.com

8 REVISION HISTORY

Rev. 0.10: 2021-09-03

- Preliminary version release

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